

Lin Ding Shan

AgriTech Computer Vision | Edge AI Deployment

Kuala Lumpur, Malaysia · Open to relocation to Australia and New Zealand

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PROFESSIONAL SUMMARY

Computer-vision engineer who builds offline vision products and ships them on edge hardware. Designed, built and field-tested a handheld durian disease and pest scouting device on Raspberry Pi 5 end to end: data pipeline, model training, ONNX export, quantisation, and on-device latency, power and thermal engineering. Halved response time and fixed field reliability issues without changing hardware. BSc Data Science, graduating May 2027.

TECHNICAL SKILLS

Languages: Python, SQL, Bash

Computer vision: PyTorch, Ultralytics YOLO (v8, 11), RT-DETR, Faster R-CNN, MobileNet, EfficientNet, OpenCV

Edge deployment: Raspberry Pi 5, Jetson Orin Nano (CUDA), Linux, ONNX Runtime, LiteRT / TFLite, INT8 and dynamic quantisation, Docker, libcamera

Performance engineering: latency, power and thermal telemetry, battery / UPS telemetry over I2C, hardware counters, duty cycling, multi-model scheduling

Data pipeline: field data collection, annotation protocols, dataset versioning, evaluation on unseen sites

Tools: Git/GitHub, Flask, AWS, Telegram Bot API, Matplotlib

EXPERIENCE

Founder & Lead Engineer | Durian edge-AI scouting device (pre-incorporation agri-tech venture), Malaysia | 2025 – Present

- Designed and built a battery-powered handheld device on Raspberry Pi 5: a grower photographs a leaf or trunk and gets a suggestion across 12 disease and pest categories in ~350 ms, fully offline. Owned capture, ONNX inference, decoding and display on one board; field-tested prototype, grower trials next.
- Owned the data pipeline: ~1,300 field images from 11 orchard sites in Peninsular Malaysia and Sabah, an annotation protocol set with a crop-protection practitioner, and evaluation on orchards held out from training.
- Halved response time (718 ms to 351 ms) by replacing two specialist detectors with one unified model after on-device profiling showed no measurable accuracy difference; cache-miss traffic also halved.
- Screened quantisation on four backbones: INT8 dropped the two smallest pest categories on every backbone while aggregate accuracy looked fine, so shipped weight-only quantisation: FP32-level accuracy at 24–40% lower energy per image.
- Made the device field-reliable: traced interruptions to the 82 °C thermal cut-off and removed them with a 30 s duty-cycle sleep; cut peak temperature by 12.7 °C and pest-model latency by 33% by running the two detectors sequentially.
- Measured where the battery goes: a runtime switch that pauses inference while capture keeps running showed 6.3 W between inferences against 8.4 W inferring, so model optimisation can save at most about half the energy; 1.6 W of the idle draw was the camera loop, making capture rate the next lever.
- Evaluated a premium Jetson Orin Nano Super tier: ported the pipeline (ONNX Runtime CUDA, Docker) and benchmarked both boards on battery, 5.4× faster pest detection (186 to 35 ms) at 2.9× cost and ~1.4× power. Dropped it because the board needed repeated OS reflashes, which a grower cannot do in the field; Pi 5 already met the response-time target.

AI Development Intern | Urban Farm Tech, Malaysia | Jan 2026 – May 2026

- Delivered an offline basil disease-monitoring node for a commercial greenhouse: ~1,700 on-site images, MobileNetV2 exported to ONNX on Raspberry Pi 5, Telegram alerts confirmed over 3-of-5 frames, under 2.5 s end to end.
- Benchmarked native vs Docker deployment over four interleaved 3-hour runs with power telemetry: the container's apparent power saving vanished per inference and its 46% latency gap came from an ONNX Runtime version mismatch; wrote an environment-manifest tool so deployments stay reproducible.

PUBLICATIONS

Sole-author paper on thermal and duty-cycle scheduling for edge inference accepted at Smart Agricultural Technology (Elsevier); six further manuscripts under review at IEEE Transactions and Elsevier journals.

EDUCATION

Bachelor of Computer Science in Data Science with Honours | UCSI University, Kuala Lumpur | Sep 2024 – May 2027 (expected)

Languages: English (fluent), Chinese (native)